

Thermodynamic State Vector Discrepancy Absolute Difference Math Function in Planetary Metabolism Simulations

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Sprint 071

Abstract

In computational models of planetary metabolism and biogeochemical cycles, maintaining rigorous adherence to physical conservation laws is paramount. Sprint 071 introduces a foundational pure helper function, `computeAbsoluteStockDelta(actual, expected)`, implemented within `src/thermodynamics/state_validator.ts`. This utility computes element-wise absolute discrepancies between measured actual thermodynamic state vectors and expected baseline vectors across five core elemental and energetic dimensions (Carbon, Nitrogen, Phosphorus, Water, and Energy). By decoupling numerical difference logic from validation state checking, the Web of Life architecture reinforces its verification layers against First Law mass-energy conservation violations and Second Law entropy management discrepancies.

1 Introduction and Systems Ecology Context

The Web of Life simulation models planetary metabolism through discrete thermodynamic state vectors and monad stock transformations. As systems evolve via bio-geochemical cycles (`src/cycles/`), validators must continuously verify that state deviations remain bounded and accounted for. Unmonitored mass generation or unexplained energy dissipation represent structural flaws that must be captured at the monad reduction boundary.

Sprint 071 formalizes the mathematical and programmatic mechanisms required to evaluate thermodynamic residuals. By isolating the delta calculation into a pure function, the system ensures deterministic, side-effect-free evaluations across all spatial nodes and ecosystem iterations. Official repository reference: <https://github.com/pascalranoroarijaona/WebOfLife>.

2 Mathematical Formalization of Elemental Stock Vectors

Let a thermodynamic state vector $\mathbf{S}$ be defined as a mapping over the set of elemental and energetic keys $\mathcal{K}$:

$$\mathcal{K} = \{\text{carbon, nitrogen, phosphorus, water, energy}\}$$

For any given spatial node or monad process i , the state vector is represented as:

$$\mathbf{S}_i = \begin{pmatrix} S_{i,\text{carbon}} \\ S_{i,\text{nitrogen}} \\ S_{i,\text{phosphorus}} \\ S_{i,\text{water}} \\ S_{i,\text{energy}} \end{pmatrix}$$

2.1 Stock Transfer & Delta Equation

Given an actual measured state vector $\mathbf{S}_{\text{actual}}$ resulting from a monad execution and an expected baseline state vector $\mathbf{S}_{\text{expected}}$, the absolute discrepancy vector Δ_{abs} is computed element-wise for each key $k \in \mathcal{K}$:

$$\Delta_{\text{abs},k} = |S_{\text{actual},k} - S_{\text{expected},k}|$$

Where any uninitialized or missing stock key defaults to the zero-state boundary:

$$S_{\text{actual},k} \leftarrow S_{\text{actual},k} \vee 0, \quad S_{\text{expected},k} \leftarrow S_{\text{expected},k} \vee 0$$

3 Implementation (src/thermodynamics/state_validator.ts)

The pure calculation function is implemented with strict type safety and fallback handling:

```
export type ElementalStockKey = 'carbon' | 'nitrogen' | 'phosphorus' | 'water' | 'energy';

export type ThermodynamicStockMap = Record<ElementalStockKey, number>;

/**
 * Computes the absolute stock discrepancy between actual and expected thermodynamic states
 * Enforces First Law residual checks by mapping exact absolute deltas per elemental key.
 */
export function computeAbsoluteStockDelta(
  actual: ThermodynamicStockMap,
  expected: ThermodynamicStockMap
): ThermodynamicStockMap {
  const result = {} as ThermodynamicStockMap;
  const keys: ElementalStockKey[] = ['carbon', 'nitrogen', 'phosphorus', 'water', 'energy'];

  for (const key of keys) {
    const actVal = actual[key] ?? 0;
    const expVal = expected[key] ?? 0;
    result[key] = Math.abs(actVal - expVal);
  }

  return result;
}
```

4 Thermodynamic Compliance & Verification

- **First Law (Mass Conservation):** Absolute differences computed by `computeAbsoluteStockDelta` serve as residuals for mass balance checks. If $\sum_{k \in \text{MassKeys}} \Delta_{\text{abs},k} > \epsilon$ (where ϵ is machine precision tolerance), a closed-system boundary leak or unmonitored mass flux has occurred.
- **Second Law (Entropy & Dissipation):** The absolute energy delta $\Delta_{\text{abs, energy}}$ isolates uncounted work, metabolic heat dissipation, or internal energy generation anomalies, feeding directly into Second Law entropy tracking routines (`src/thermodynamics/monad_process.ts`).

5 Conclusion

Sprint 071 establishes a robust, pure mathematical foundation for state validation within the Web of Life architecture. By standardizing discrepancy calculations across elemental and energetic stock maps, the framework ensures absolute traceability of mass-energy transformations across planetary metabolic simulations.